olmoearth_inferenceX: examining OlmoEarth inference results without labels

which windows of a prediction map to trust, and which to send for review

A What is audited

Sentinel-2 window

OlmoEarth encoder

v1 Base, frozen, fp32

three deployment cases

a linear water probe on frozen tokens
(27 WorldCover scenes; Sen1Floods11 tiles)
the fine-tuned AWF model end to end
the served LCC change rasters

prediction map + per-window confidence

errors = disagreements with a reference, patch by patch; the signal's job is to rank them before anyone looks

references: WorldCover (weak map), Sen1Floods11 hand labels, AWF expert points

B What we believe is true

- **the model's own confidence is the best label-free error ranker**
  negative logit margin; best on every expert-labelled testbed (AWF, Sen1Floods11, fine-tuned model)
  exp04 16 18 21
- **errors concentrate on prediction boundaries: 75% of errors vs 20% of correct**
  a triage cue, not a ranker; boundary first, then by confidence, captures more errors at 5-10% budgets
  exp14 16 18 20 35 36
- **an error window nearly always carries a label-free cue**
  95% of hand-label errors are on a boundary, low-confidence, unstable or NDWI-ambiguous; NDWI ambiguity 7x
  exp37
- **the fine-tuned model is overconfident, so an accuracy needs a coverage**
  0.93 accurate where it claims 0.99 (ECE 0.08); abstaining on the least confident 20% gives 0.945
  exp21
- **the served product exports no class confidence; outputs sit on the patch lattice**
  boundary fraction captures a median 0.88 of disagreements at a 5% review budget; no window seams
  exp20 22
- **side product: the pretraining target space is degenerate**
  target encoder = untouched random init, targets of effective rank 2; whitened, the target is 57-70% predictable
  exp32 33 34

every item: supported on expert labels; what was tried and rejected is in the ledger (docs/TECHNIQUES.md)

C The test every claim passed

two references at once, on identical patches

the model's own confidence and a no-model pixel control
beating one but not the other is not support

scoring

tie-aware excess AURC per scene or tile
wins / losses / ties, exact sign tests, sign-flip permutation
one vote per river cluster; block and cluster bootstraps
preregistered primary score and combination

rules

expert labels grade signals, never train them
every recorded claim points to a file under exp/out
the machinery is a tested torch-free package

D Open question

why the WorldCover wins do not transfer to hand labels.
leading hypothesis: WorldCover was a pretraining target, so the probe partly reads out the model's own map; decisive test needs adjudicated cells on the 8 rivers (issue #2)

E Where it stands

supported for deployment: confidence as the ranker, boundary first then confidence as the review order, and a reason with measured evidence for every flagged window (oe_inferencex).
Remaining: the adjudicated-cell test (issue #2) and the pretraining-target note to Ai2 (issue #11)

github.com/2imi9/olmoearth_inferenceX; ledger docs/TECHNIQUES.md; protocol docs/method/protocol.md; open items docs/plan/roadmap.md